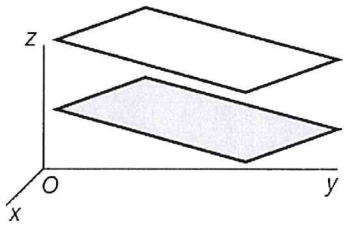
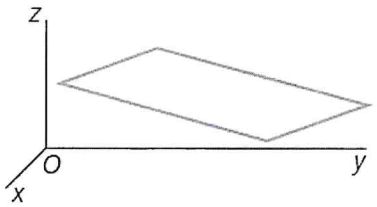
Intersection of Two Planes

Two planes can intersect in the following ways:



Planes are PARALLEL & DISTINCT

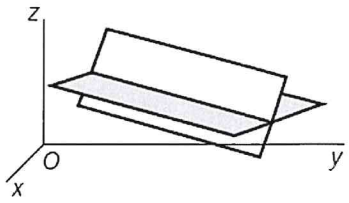
NO Solutions



Planes are COINCIDENT

Infinite # of Solution

Solution is the entire plane.



Planes INTERSECT

Infinite # Solution

Solution is a Line.

$\vec{n_1} \not\parallel \vec{n_2}$

$\therefore \vec{n_1} = k\vec{n_2}$

Example 1: Determine the intersection of the planes

$$x + 4y - 3z = 6$$
$$2x + 8y - 6z = -11$$

$\vec{n_1} = (1, 4, -3)$
$\vec{n_2} = (2, 8, -6)$
} collinear

Option # 1

Since $2\vec{n_1} = \vec{n_2}$

$\therefore$ parallel planes.

· check coincident:

$(① \; x + 4y - 3z = 6) \times 2$

① $2x + 8y - 6z = 12$

② $2x + 8y - 6z = -11$

not the same

$\therefore$ planes are $\not\parallel$ and distinct.

$\therefore$ no solutions

Option #2

$$\left[\begin{array}{ccc|c} 1 & 4 & -3 & 6 \\ 2 & 8 & -6 & -11 \end{array}\right] \rightarrow 2R_1 - R_2 \left[\begin{array}{ccc|c} 1 & 4 & -3 & 6 \\ 0 & 0 & 0 & 23 \end{array}\right]$$

By R2 : $0z = 23$.

not possible.

$\therefore$ no solutions.

Challenge: What's the **ONE** change we can make to the system above so that there will be infinite solutions?

a) Coincident Planes? same constant ② $2x + 8y - 6z = 12$

b) Intersecting Planes? non-collinear normals.

② $2x + 7y - 6z = -11$

↑ answers could vary

Example 2: Determine the intersection of the planes

$2x+3y+4z=10$
$3x+y-z=1$

$\vec{n_1}=(2,3,4)$
$\vec{n_2}=(3,1,-1)$

$\vec{n_1} \not\parallel \vec{n_2}$ ∴ Intersect in a line!

$$\left[\begin{array}{ccc|c} 2 & 3 & 4 & 10 \\ 3 & 1 & -1 & 1 \end{array}\right] \rightarrow_{3R1-2R2} \left[\begin{array}{ccc|c} 2 & 3 & 4 & 10 \\ 0 & 7 & 14 & 28 \end{array}\right] \rightarrow_{\frac{1}{7}R2} \left[\begin{array}{ccc|c} 2 & 3 & 4 & 10 \\ 0 & 1 & 2 & 4 \end{array}\right]$$

target ↙ (3)

target to eliminate →

$$\xrightarrow{R_1-3R2} \left[\begin{array}{ccc|c} 2 & 0 & -2 & -2 \\ 0 & 1 & 2 & 4 \end{array}\right] \rightarrow_{\frac{1}{2}R_1} \left[\begin{array}{ccc|c} 1 & 0 & -1 & -1 \\ 0 & 1 & 2 & 4 \end{array}\right]$$

Reduced Row-Echelon Form or "Gauss-Jordan" Form

R_1: $x-z=-1$
$x=z-1$

R_2: $y+2z=4$
$y=-2z+4$

Let $z=t$

$$\therefore \begin{cases} x=t-1 \\ y=-2t+4 \\ z=t \end{cases} t\in\mathbb{R}$$

$$\therefore \vec{r}=(-1,4,0)+t(1,-2,1),\ t\in\mathbb{R}$$

Example 3: Determine the angle of intersection between the two planes.

$2x+3y+4z=10$
$3x+y-z=1$

$\vec{n_1}=(2,3,4)$
$\vec{n_2}=(3,1,-1)$

$$\cos\theta=\frac{(2,3,4)\cdot(3,1,-1)}{\sqrt{2^2+3^2+4^2}\sqrt{3^2+1^2+(-1)^2}}$$

$$\cos\theta=\frac{5}{\sqrt{29}\sqrt{11}}$$

$\theta=74°$

Recall:
1. Planes are parallel if:
$\vec{n_1} \not\parallel \vec{n_2}$ or $\vec{n_1}=k\vec{n_2}$
2. Planes are perpendicular if:
$\vec{n_1}\cdot\vec{n_2}=0$
3. Angle between intersecting planes can be calculated using:
$$\cos\theta=\frac{\vec{n_1}\cdot\vec{n_2}}{|\vec{n_1}||\vec{n_2}|}$$